Groundwater Geostatistics and Inverse Modeling

A teaching guide with mathematical derivations, questions, solutions, and a pyRGA case study.

> Audience: students with basic calculus, linear algebra, probability, and introductory groundwater hydrology.
> Scope: variograms → Gaussian random fields → kriging → co-kriging → groundwater inversion → optimization.

0. The case study and roadmap

Consider an aquifer with heterogeneous positive hydraulic conductivity $K(\mathbf{x})$. We may have sparse borehole measurements of K, and head observations from multiple pumping configurations. We want to (1) characterize spatial continuity, (2) estimate unsampled properties, and (3) infer a conductivity field consistent with observations and groundwater physics.

Method	Question	Result
Variogram	How does dissimilarity change with separation?	Spatial dependence model
Gaussian field	What heterogeneous fields are plausible?	Statistical prior/realizations
Kriging	What is $\log K$ at unsampled points?	Best linear prediction and variance
Co-kriging	Can auxiliary observations improve predictions?	Multivariate spatial prediction
Inversion	Which conductivity fields explain observed heads?	Calibrated parameter field
Optimization	How do we efficiently solve the inverse problem?	Parameter estimates and diagnostics

Why use $Y = \log K$? Conductivity must be positive, and the logarithm often gives a more manageable distribution. A Gaussian model for Y produces positive, lognormal $K = \exp(Y)$. The lognormal assumption is a model choice, not a universal law.

1. Variograms and covariance

1.1 Random-field statistics

For a spatial process $Y(\mathbf{x})$, define

$$m(\mathbf{x}) = E[Y(\mathbf{x})], \qquad C(\mathbf{x}, \mathbf{x}') = \mathrm{Cov}(Y(\mathbf{x}), Y(\mathbf{x}')).$$

Second-order stationarity means a constant mean and covariance depending only on the separation vector $\mathbf{h} = \mathbf{x}' - \mathbf{x}$. Isotropy additionally means covariance depends only on $h = \|\mathbf{h}\|$.

1.2 Semivariogram

$$\gamma(\mathbf{h}) = \frac{1}{2} E[(Y(\mathbf{x}+\mathbf{h}) - Y(\mathbf{x}))^2].$$

Under second-order stationarity, expand the square:

$$2\gamma(\mathbf{h}) = \mathrm{Var}(Y_1) + \mathrm{Var}(Y_2) - 2\,\mathrm{Cov}(Y_1, Y_2) = 2C(0) - 2C(\mathbf{h}),$$

so

$$\boxed{\gamma(\mathbf{h}) = C(0) - C(\mathbf{h})}.$$

Estimate an isotropic semivariogram from sampled pairs in a lag bin $B(h)$:

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{(i,j)\in B(h)} [Y(\mathbf{x}_i) - Y(\mathbf{x}_j)]^2.$$

The **nugget** is a discontinuity at zero distance (microscale variability or measurement error); **sill** is the large-distance semivariance of a stationary process; **range** measures spatial continuity. An exponential model approaches its sill asymptotically, so distinguish the scale parameter from the **practical range**.

1.3 Three models

For $h > 0$, with nugget c_0, partial sill c:

- Exponential: $\gamma(h) = c_0 + c(1 - e^{-h/a})$; practical range for 95% of partial sill $\approx 3a$.
- Gaussian: $\gamma(h) = c_0 + c(1 - e^{-(h/a)^2})$; practical range $\approx \sqrt{-\log(0.05)}a \approx 1.73a$.
- Spherical: $\gamma(h) = c_0 + c[1.5(h/a) - 0.5(h/a)^3]$ for $0 < h \le a$, and $c_0 + c$ beyond a.

Theoretical $\gamma(0) = 0$ even with a nugget; $\lim_{h\downarrow 0} \gamma(h) = c_0$.

Anisotropy may be expressed by $r = \sqrt{(\Delta x/a_x)^2 + (\Delta y/a_y)^2}$ and $C(\Delta x, \Delta y) = \sigma^2 e^{-r}$, with direction-specific correlation scales. Statistical anisotropy of the log-K field is different from a physical anisotropic conductivity tensor.

Worked problem 1

Given $C(h) = 2e^{-h/10}$, calculate the semivariogram at $h = 10$.

Solution: $\gamma(10) = 2(1 - e^{-1}) \approx 1.2642$.

Questions with answers

1. **Why do nearby observations tend to be more similar?** When covariance decreases with separation, near points are more positively correlated; geological discontinuities can violate this simple expectation.
2. **Does a 100 m variogram range imply constant K inside 100 m?** No. It describes decay of statistical dependence, not a homogeneous block.
3. **Why might an empirical variogram exceed the fitted sill?** Sampling noise, nonstationarity, a wrong model, directional mixing, or outliers.
4. **What changes when the field is anisotropic?** Dependence is a function of the *separation vector* rather than distance alone; compute directional variograms.

2. Gaussian random fields (GRFs)

For finite points $\mathbf{x}_1, \ldots, \mathbf{x}_n$, a GRF satisfies

$$\mathbf{Y} = [Y(\mathbf{x}_1), \ldots, Y(\mathbf{x}_n)]^T \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}).$$

A valid covariance matrix must be symmetric and positive semidefinite for every set of locations. A field having a *Gaussian-shaped covariance kernel* does **not** by itself imply a Gaussian probability distribution.

2.1 Sampling and conditioning

With $\mathbf{C} = \mathbf{L}\mathbf{L}^T$ and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$,

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}.$$

For unobserved u and observed o components of a joint Gaussian vector,

$$\boxed{E[\mathbf{Y}_u \mid \mathbf{y}_o] = \boldsymbol{\mu}_u + \mathbf{C}_{uo}\mathbf{C}_{oo}^{-1}(\mathbf{y}_o - \boldsymbol{\mu}_o)}$$

$$\boxed{\mathbf{C}_{u|o} = \mathbf{C}_{uu} - \mathbf{C}_{uo}\mathbf{C}_{oo}^{-1}\mathbf{C}_{ou}}.$$

If observations have independent additive error with covariance $\mathbf{R}$, replace $\mathbf{C}_{oo}$ by $\mathbf{C}_{oo} + \mathbf{R}$ when conditioning on noisy data.

2.2 Reduced-dimensional parameterization

A truncated Karhunen–Loève/principal-component-style representation is

$$Y(\mathbf{x}, \boldsymbol{\xi}) \approx m(\mathbf{x}) + \sum_{j=1}^{r} \sqrt{\lambda_j}\phi_j(\mathbf{x})\xi_j,$$

where $\xi_j \sim \mathcal{N}(0,1)$ under the Gaussian prior. Instead of estimating thousands of cellwise values, estimate the retained coefficients. The precise scaling and basis construction in a particular software package must be checked rather than assumed identical to this expression.

Worked problem 2

Let standardized Gaussian Y_1, Y_2 have correlation $\rho = 0.8$. After observing $Y_1 = 2$, what are $E[Y_2 \mid Y_1]$ and $\mathrm{Var}(Y_2 \mid Y_1)$?

Solution: conditional mean $0.8(2) = 1.6$; conditional variance $1 - 0.8^2 = 0.36$.

Questions with answers

1. **Can two fields have the same covariance but different realizations?** Yes; covariance describes ensemble statistics, not a unique field.
2. **Does conditioning always make every field value exact?** Only at noise-free directly observed points under an exact conditioning model.
3. **Why use a reduced basis?** It reduces inverse problem dimension and can encode prior smoothness/correlation.
4. **Is the conditional mean a typical realization?** Not necessarily. The mean suppresses unresolved high-frequency variability.

3. Kriging

Write a linear estimate at $\mathbf{x}_0$ as $\hat{Y}_0 = \sum_i w_i Y_i$ (ordinary kriging), or $\hat{Y}_0 = m + \sum_i w_i (Y_i - m)$ for known mean m (simple kriging).

3.1 Simple kriging

$$\boxed{\hat{Y}_0 = m + \mathbf{c}^T \mathbf{C}^{-1} (\mathbf{y} - m\mathbf{1})}$$

$$\boxed{\sigma^2_{SK} = C(0) - \mathbf{c}^T \mathbf{C}^{-1} \mathbf{c}},$$

with $\mathbf{c}_i = \mathrm{Cov}(Y(\mathbf{x}_0), Y(\mathbf{x}_i))$. Under a joint Gaussian model, these equal the conditional mean and variance.

3.2 Ordinary kriging

Unknown constant mean requires unbiasedness $\mathbf{1}^T \mathbf{w} = 1$:

$$\begin{bmatrix} \mathbf{C} & \mathbf{1} \\ \mathbf{1}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \eta \end{bmatrix} = \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}.$$

With this sign convention, $\sigma^2_{OK} = C(0) - \mathbf{w}^T \mathbf{c} - \eta$.

Example: At the midpoint of two symmetrically located wells with values 10 and 14, ordinary kriging weights are 0.5 and 0.5, producing 12. The numerical prediction *variance* still requires a covariance model.

Practical distinction: Kriging from borehole conductivity data is *not* the same as assimilating hydraulic heads through a groundwater PDE. In our code, kriging uses synthetic **direct log-conductivity samples**; inversion uses **heads**.

Questions with answers

1. **Can a kriging weight be negative?** Yes; weights are constrained covariance-optimal coefficients, not probabilities.
2. **What makes ordinary and simple kriging different?** Simple kriging treats the mean as known; ordinary kriging accounts for an unknown constant mean.
3. **What uncertainty does reported kriging variance omit?** Commonly, uncertainty in the fitted variogram parameters and structural misspecification.
4. **Why is a kriging map often too smooth?** It is a conditional mean rather than a conditional simulation.

4. Co-kriging and secondary data

Let $Y_1 = \log K$ (primary) and Y_2 (e.g., geophysical measurement) be two jointly modeled fields. Use the covariance blocks

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}.$$

Known-mean simple co-kriging is joint Gaussian conditioning:

$$\hat{Y}_{1,0} = m_{1,0} + \mathbf{C}_{1,0,d} \mathbf{C}_{dd}^{-1} (\mathbf{d} - \mathbf{m}_d).$$

For ordinary co-kriging, apply appropriate mean/unbiasedness constraints. Cross-covariances cannot be fitted arbitrarily: the *whole* multivariate covariance must remain valid.

A valid **linear model of coregionalization (LMC)** is

$$C_{ij}(\mathbf{h}) = \sum_k B_{ij}^{(k)} \rho_k(\mathbf{h}),$$

where ρ_k are valid spatial correlations and each $\mathbf{B}^{(k)}$ is positive semidefinite.

Example: Borehole $\log K$ is sparse but geophysical resistivity is dense. Co-kriging may improve inference if joint correlation is supported. Be careful: resistivity is also influenced by pore-water chemistry, lithology, and saturation.

Questions with answers

1. **When does a secondary variable help?** When it provides nonredundant, reliable information about the primary variable under a valid joint model.
2. **Why can't we fit arbitrary cross-variograms separately?** The assembled joint covariance may cease to be positive semidefinite.
3. **How does regression kriging differ?** It uses predictors for the mean trend and kriges residuals, whereas co-kriging models cross-spatial dependence explicitly.
4. **Does `pyrga-ht` provide a co-kriging demonstration in this repository?** No; this repository presents co-kriging theoretically, while the runnable lab implements a univariate kriging step and an RGA head inversion.

5. Groundwater forward and inverse modeling

5.1 Governing equation

For saturated transient flow with scalar K (or tensor $\mathbf{K}$), a common continuum formulation is

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (\mathbf{K} \nabla h) + q,$$

where S_s is specific storage and q is a consistent volumetric source density. For steady conditions, $\nabla \cdot (\mathbf{K} \nabla h) + q = 0$. Boundary conditions, geometry, units, and source normalization matter.

Let $\mathbf{m} = \log \mathbf{K}$ and let $\mathcal{G}(\mathbf{m})$ map parameters to measured heads (including numerical flow solution and observation extraction). Then

$$\mathbf{d} = \mathcal{G}(\mathbf{m}) + \boldsymbol{\epsilon}.$$

5.2 Misfit and regularization

A common inverse objective is

$$J(\mathbf{m}) = \frac{1}{2} \| \mathcal{G}(\mathbf{m}) - \mathbf{d} \|^2_{\mathbf{R}^{-1}} + \frac{1}{2} \| \mathbf{m} - \mathbf{m}_0 \|^2_{\mathbf{C}_m^{-1}}.$$

Here $\mathbf{R}$ is measurement-error covariance and $\mathbf{C}_m$ is prior covariance. Under additive Gaussian errors and Gaussian prior, this is the negative log-posterior up to constants; minimization yields a **maximum a posteriori (MAP)** estimate.

Sparse head observations rarely identify thousands of independent conductivities. Spatial priors, parameter reduction, pumping experiments, and additional flux/tracer data can reduce—but rarely eliminate—ambiguity.

5.3 Linear-Gaussian result

For $\mathbf{d} = \mathbf{H}\mathbf{m} + \boldsymbol{\epsilon}$ with independent Gaussian prior and errors,

$$\mathbf{C}_{post} = (\mathbf{C}_m^{-1} + \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H})^{-1},$$

$$\mathbf{m}_{post} = \mathbf{m}_0 + \mathbf{C}_m\mathbf{H}^T(\mathbf{H}\mathbf{C}_m\mathbf{H}^T + \mathbf{R})^{-1}(\mathbf{d} - \mathbf{H}\mathbf{m}_0).$$

This is connected to Gaussian conditioning, kriging, and Kalman updates. A nonlinear groundwater posterior need not be Gaussian, and its MAP need not equal its mean.

5.4 What makes hydraulic tomography informative?

Changing which well pumps changes the head sensitivity patterns. Multiple source/observation configurations probe different aspects of subsurface structure. Nonetheless, parameter null spaces remain possible and model-error effects should be assessed.

Questions with answers

1. **Why can excellent head fit coexist with poor conductivity recovery?** Nonuniqueness, noise, model error, sparse coverage, and regularization bias.
2. **Why use log-K parameters?** Positivity and often improved statistical scaling.
3. **How is kriging distinct from inversion?** Kriging uses covariance and direct spatial measurements; inversion incorporates a PDE connecting K to heads.
4. **What is the role of the prior?** It constrains plausible magnitudes, correlations, and structures that head data cannot uniquely determine.

6. Optimization

For residual $\mathbf{r}(\mathbf{m}) = \mathcal{G}(\mathbf{m}) - \mathbf{d}$ and sensitivity Jacobian $\mathbf{J}_G = \partial\mathcal{G}/\partial\mathbf{m}$, Gauss–Newton solves an approximate second-order system. For the MAP objective above,

$$\left(\mathbf{J}_G^T\mathbf{R}^{-1}\mathbf{J}_G + \mathbf{C}_m^{-1}\right)\Delta\mathbf{m} = -\mathbf{J}_G^T\mathbf{R}^{-1}\mathbf{r} - \mathbf{C}_m^{-1}(\mathbf{m} - \mathbf{m}_0).$$

The update $\mathbf{m} \leftarrow \mathbf{m} + \Delta\mathbf{m}$ may require line search/trust region. Levenberg–Marquardt adds damping; BFGS approximates curvature; adjoints efficiently compute derivatives with respect to many parameters.

For discretized $\mathbf{A}(\mathbf{m})\mathbf{h} = \mathbf{b}(\mathbf{m})$, define the adjoint by $\mathbf{A}^T\boldsymbol{\lambda} = (\partial J/\partial\mathbf{h})^T$. Then

$$\nabla_{\mathbf{m}} J = \left(\frac{\partial J}{\partial \mathbf{m}}\right)_{\mathbf{h}} + \left[\frac{\partial \mathbf{b}}{\partial \mathbf{m}} - \frac{\partial(\mathbf{A}\mathbf{h})}{\partial \mathbf{m}}\right]^T \boldsymbol{\lambda}.$$

A scalar objective can require one adjoint solve rather than a forward solve per unknown parameter, subject to implementation details.

Questions with answers

1. **Why is Gauss–Newton attractive for head-data fitting?** It exploits residual sensitivities and least-squares structure.

2. **What does the Jacobian mean physically?** Each element describes the sensitivity of one predicted observation to one parameter.
3. **Does optimization guarantee a physically correct field?** No. It optimizes the chosen objective; identifiability and model adequacy must also be checked.
4. **Why use regularization?** To stabilize ill-posed estimation and encode prior plausibility.

7. pyRGA hands-on lab

Run the accompanying Python demo after following README.md. It follows the PyPI `pyrga-ht` v0.1.4 public example interface:

1. `generate_synthetic_field`: create a positive heterogeneous K and reduced spatial basis.
2. Empirical isotropic semivariogram from sampled pairs of $\log K$.
3. Simple kriging on a subset of **synthetic direct conductivity samples** using a **separately chosen illustrative Gaussian covariance** (not claimed to be pyRGA's exact generator covariance).
4. `prepare_physical_domain` and `hydraulic_tomography`: simulate pumping-well flow responses.
5. `observation_operator` and `add_noise`: construct noisy head observations.
6. `forward_model` and `gauss_newton_dynamic_lambda`: estimate reduced coefficients through the package's RGA optimization.
7. Diagnostic figures and result archive for comparison.

Important limits: This is a synthetic tutorial, not calibration of real site data. The exact flattening and units should be checked against pyRGA implementation. The demo evaluates **observation-space fit**; do not claim accurate recovered spatial K solely from head RMSE. Co-kriging is discussed in Section 4 but is not implemented in the demo.

Experiments

- Change the field's `lx` and `ly` and compare structure and sample variograms.
- Increase/decrease synthetic borehole count and plot kriging standard deviation.
- Compare kriging error against the synthetic field in log space; repeat across seeds.
- Change the observation noise level and examine inversion error and stability.
- Change the number of retained modes and consider representation error versus conditioning.
- Contrast initial and fitted head predictions; analyze where model predictions are biased.

Lab questions with model answers

L1. Does a low kriging prediction variance mean the covariance model is correct? No. It reports uncertainty conditional on an assumed covariance and data geometry.

L2. Why is the sample variogram noisy? One realization, limited sampled pairs, lag binning, and statistical variability.

L3. Does optimizing reduced coefficients use direct K samples? Not in the pyRGA inversion step used here; it uses hydraulic-head observations, while the kriging exercise uses separately sampled log-K values.

L4. Why might inversion heads fit better without reconstructing the true field? Multiple fields can generate similar response patterns; reducing basis dimension adds approximation bias.

L5. Does decreasing the regularization coefficient always improve the result? It may improve in-sample fit but harm parameter plausibility or out-of-sample prediction.

8. Ten-question final assessment (with solutions)

1. **Derive $\gamma(h) = C(0) - C(h)$.** Expand the squared two-point difference and use second-order stationarity.
2. **For $C(h) = 2e^{-h/10}$, find $\gamma(10)$.** $2(1 - e^{-1}) \approx 1.2642$.
3. **What makes a covariance valid?** Every finite covariance matrix must be symmetric positive semidefinite.
4. **Why is the kriging mean smoother than a field realization?** It integrates out residual conditional uncertainty.
5. **What changes under observation noise?** Gaussian conditioning uses $C_{oo} + R$ for independent additive noise.
6. **What constraint matters for cross-covariance?** The full joint covariance must remain valid/positive semidefinite.
7. **Define the inverse problem.** Estimate K or reduced parameters that explain observed heads through the forward groundwater model.
8. **When does quadratic regularization equal Gaussian-prior MAP?** With Gaussian errors and a Gaussian prior (properly weighted).
9. **How can head fit be perfect but K wrong?** Nonidentifiability and limited observations.
10. **When are adjoints useful?** Many parameters and few scalar objectives, when derivatives of a PDE-based objective are required.

9. Suggested four-week teaching plan

Week	Focus	Assignment
1	Variogram, stationarity, anisotropy, GRFs	Generate fields and calculate directional/omnidirectional variograms
2	Conditional Gaussian, kriging, co-kriging	Run kriging and interpret standard deviation; derive simple co-kriging equations
3	Flow PDE, observation operator, inversions	Run pyRGA tomography and explain observation sensitivities
4	RGA, Gauss–Newton, regularization	Complete inversion, analyze fit and identifiability, present limitations

References and attribution

- pyRGA package / published v0.1.4 example. API function calls are adapted from its public quick-start/optimization example; this repository's explanatory code and exercises are separately authored.
- Zhao, Y., & Luo, J. (2020), *Reformulation of Bayesian Geostatistical Approach on Principal Components*, **Water Resources Research**, 56(4), e2019WR026732, https://doi.org/10.1029/2019WR026732 .
- Cressie, N., *Statistics for Spatial Data*.
- Chilès, J.-P., & Delfiner, P., *Geostatistics: Modeling Spatial Uncertainty*.
- Anderson, M. P., Woessner, W. W., & Hunt, R. J., *Applied Groundwater Modeling*.